

Project brief

Shamsy all-in-one business platform · confidential · for shortlisted candidates only

1 · What this business actually does

Shamsy imports solar equipment and sells it in Sudan. Inverters, batteries, solar air conditioners, and from this year solar panels, balance-of-system parts and solar water pumps. We buy from suppliers in Saudi Arabia, China and the UAE. We do not install anything. We sell to roughly 570 dealers and installers who do the installing, and to a smaller number of large end customers.

A shipment arrives in Port Sudan, clears customs, and is trucked to our warehouse in Omdurman. From there it goes out in ones and twos to dealers across the country. Our latest battery and inverter container was over \$220,000 of goods at purchase price; a container of panels is 720 panels, around \$93,000 in revenue.

The war destroyed most of the national grid. Generating capacity fell from 4,400 MW to about 1,100 MW, and blackouts run to eighteen hours a day. Solar is how people keep a freezer running and a phone charged. Demand is not our problem. Cash, currency and coordination are.

2 · How the money moves — read this before anything else

The whole system exists because Sudan's banking does not work normally.

Euros → dollars → stock → Sudanese pounds → euros. The owner starts with euros. Suppliers want dollars, so he exchanges euros for dollars and pays a commission on top. Worked back, that tells him what each dollar of stock cost him in euros: the **cost factor**.

On the August 2026 shipment he received 1.1674 dollars per euro and paid €211 commission. So every dollar of goods cost him €0.860234. Check: \$59,983 of goods × 0.860234 = €51,599, almost exactly the €51,593 he put into that shipment.

Each shipment gets its own factor, and once set it never changes: the euros were spent on the day of the exchange, so a later move in the euro-dollar rate does not change what that stock cost. When a supplier is paid with dollars already held — for instance from sales received on the UAE account — the factor is what those dollars cost. The system multiplies dollar cost by the shipment's factor to get cost in euros, which is how net profit in euros is calculated.

The stock is sold in Sudan in **Sudanese pounds**. Customers pay directly into bank accounts held by **money exchangers**. The exchangers convert the pounds and pay the owner out in euros, at their rate of that day.

Every result in both currencies, as an amount and as a margin. The owner must be able to see each line below in dollars and in euros, side by side or with one switch, and each as an amount and as a percentage of revenue:

Line	What it means	In \$	In €
Gross profit	Revenue at the order rate minus landed cost. Whether we buy and sell well.	amount and margin %	amount and margin %
Net profit before currency	Gross profit minus expenses such as salaries, warehouse and transport. The business on its own merits.	amount and margin %	amount and margin %
Currency result	What the Sudanese pounds actually became, against what they were priced at. Gain or loss from the conversion.	amount	amount
Net profit after currency	Net profit plus the currency result. What the business really earned after converting.	amount and margin %	amount and margin %

In dollars, costs are the landed cost in dollars. In euros, costs are converted through the shipment's cost factor, and revenue through the rate at which the pounds were actually converted — or, while still unconverted, the rate at which they were booked.

The result depends on where and how much is converted. Pounds can be converted by an exchanger in Sudan, who pays out in euros; or paid by customers into the UAE account, where they arrive as dirham or dollars and are converted through Airwallex — or not converted at all, because dollars on the UAE account can pay the next supplier directly. With the pound falling fast, more and more payments are expected to arrive in the UAE. So the system shows the currency result per route — each exchanger, and the UAE account — with the rate achieved, the volume converted, and what it earned or cost against the pricing rate. Money still held in dollars or dirham is shown as held, at the rate it came in at, not counted as euro profit until it is actually converted.

Prices are set in dollars. Each morning the day's rate is multiplied in, so what we charge in pounds never lags the currency. The pound falls almost daily: in June 2026 we priced at around 6,050 SDG per dollar; by September it is above 8,000. So the rate is never a constant in code. It is a setting the owner changes, often every day, and below it sits a **minimum rate**, also set by the owner, under which no adviser can quote.

We also price on a conservative **calculation rate** and convert at whatever the exchangers give that day. The difference is the currency result: on the August shipment it was worth \$4,135 — gross margin 25.7%, net margin 30.3%.

The system works in four currencies: US dollars (prices, purchasing, margin), Sudanese pounds (what customers pay), euros (what the owner starts and ends with) and UAE dirham (payments received through the UAE account). Every amount carries its currency and the rate at which it was converted.

The constraints everything comes from:

- A Sudanese bank account can receive about **15,000,000 SDG per day**. Exchangers will not hold more because of inflation.
- A single bank transfer is capped at **3,000,000 SDG**. A 15 million payment arrives as five transfers.
- So we work with **four exchangers — Yassir, Tahani, Ali and Zain — holding eleven accounts** between them. When an account is full, the next customer gets a different account number.
- Sometimes money lands on the owner's own operating account, **Umhani**, and has to be forwarded to an exchanger. That is an extra step.
- **Amira** has an account in the UAE that receives dirham payments and passes them on. It is a pass-through: it should always net to zero.
- To grow beyond what the exchangers can handle, customer payments are moving to a **company account in the UAE, run through Airwallex**, receiving dirham and dollars. From there money is converted or used to pay suppliers directly. The system must treat it as a full account alongside the exchanger accounts: receipts in AED or USD, conversions with their rate, supplier payments out of it.
- The pound loses value every week, so time between sale and conversion costs money.

Payment proofs arrive as screenshots — Bank of Khartoum (bankak) confirmations, mostly in Arabic, forwarded on WhatsApp by advisers, privately and in group chats. The same screenshot often arrives twice. Every proof carries a **unique transaction code** (مقررة لعملة), and that code is the key to the entire administration. The comment field often holds a customer name or a sequence like "Ihab 1" to "Ihab 6", which is how a split order is reconstructed.

The owner then has to **forward each proof to the exchanger** so the exchanger knows that money is his. He has caught himself forgetting. Money that was received but never forwarded sits with an exchanger who does not know it belongs to us.

3 • Who works in this system

The owner, in Europe. Buys, arranges shipping, negotiates supplier terms, manages the currency chain, sets the daily rate, approves large discounts and price overrides, and gives the team weekly direction. Works in English and Dutch.

The head of marketing, Amira, in the UAE. Must not see cost prices or margins. Runs Facebook (14,000 followers), TikTok (4,000) and the dealer WhatsApp groups (570 dealers). Also receives dirham payments. Announces shipments before they land, so she needs to know what is in stock and what is arriving.

Two sales advisers, Sana in Dongola and Marwa in Khartoum. Take orders, quote, chase payments, forward proofs. Work on phones, often on a bad connection. Their current sales tool is in Arabic; the new system starts in English, with Arabic added later. They must never see cost prices or margins.

The warehouse, Jawahir in Omdurman. Receives shipments, releases goods against paid orders, counts stock. Sees no financial data. Once took a cash payment, so cash receipts by staff must be possible.

4 · What already exists, and what it teaches

This is not a greenfield project. Four working tools already encode how the business runs. Read them before designing anything.

The sales tool — Shamsy_Verkooptool_v7.html, **version 7**. A single offline HTML file the advisers open on their phones, sent around as a WhatsApp attachment. Arabic, right-to-left. The adviser enters the day's rate, picks products from a fixed list, sets quantities, and builds a multi-line quote with totals in dollars and Sudanese pounds. Its rules are deliberate and must survive into the new system:

- **Advisers cannot change prices.** Dollar prices are fixed. They choose product, quantity and rate. The point is that nothing can go wrong in the calculation.
- **Cost prices and margins never appear.** An adviser sometimes hands her phone to a customer.
- **A minimum rate**, enforced three ways: the field turns red with a message, the add button refuses, and on leaving the field the value snaps back to the minimum. A higher rate is always allowed. In the current file the minimum is hard-coded — at 8,000 since version 7 — which is exactly what must not happen in the new system: the pound falls almost daily, from around 6,050 per dollar in June to above 8,000 in September, so the minimum has to be a setting the owner changes himself.
- **Only SDG and USD.** Saudi riyal was deliberately removed. With USD the rate field is disabled and set to 1.
- **Works offline**, large buttons for thumbs, numbers formatted with thousands separators.

Planned for it and still open: customer name and date on the quote so it can be shared as a receipt; editing quantity in an existing line; changing the minimum rate behind an owner-only code; a share or print button; and possibly a second price column for pick-up in Saudi Arabia, without transport and clearing costs.

The daily administration — Shamsy_Daily_Admin_v10.xlsx. Thirteen sheets: Read me, Customers, one sheet per adviser, Warehouse, Forwarding check, Rates, Cash and bank, Dashboard, Result, Products, Accounts, Checks. Order numbers carry a prefix per adviser, SA- and MA-. Each order gets two payment lines that fill in its order number automatically. A generated WhatsApp message tells the customer the amount, the account, and how many transfers of 3,000,000 it takes. Payment methods: BANKAK, UAE Account, Cash. Discounts are coloured by size, and above 5% the sheet refuses the entry unless the owner has marked the order as approved.

The shipment dossier — Shamsy_Zending_Growatt_v24.xlsx. Every cost of a shipment — invoice, sea freight, bank charges, customs clearing in several payments in SDG at their own rates, inland transport — spread over the products pro rata to purchase value, with a 20% safety uplift on indirect costs. On the current shipment that lifts the purchase price by 12.44%: an inverter bought at \$270 has a true cost of \$303.58. Also tracks margin per order and the supplier payment deadline.

The cash register — Shamsy_Kasregister. The event-based ledger built from 211 payment screenshots: one line per unique transaction code, from which account to which account, amount, comment, and whether it appears in the sales sheet. Plus a sheet of open questions where nothing is booked until the owner answers.

Also available: the price list PDF (English, sales prices only), the investor deck, the HTML dashboard prototype, all brand assets, and the product catalogues of our OEM lines.

5 · What a day looks like now, and why it hurts

A dealer sends a WhatsApp message: two 5 kWh batteries and an inverter. An adviser opens the sales tool, enters the rate, builds the quote, and sends a screenshot of it. She writes the order in the daily spreadsheet.

The dealer pays in two or three instalments, each capped at 3 million pounds, into accounts that may be full. He sends a screenshot of each transfer. The adviser forwards them to the owner. The owner forwards them to the exchangers. Someone matches them to orders by hand, in the evening, sometimes days later.

The warehouse releases goods on a WhatsApp message that sometimes arrives before the money does.

At the end of a shipment, 64 payment proofs are reconciled against 22 orders across three currencies. The earlier ledger was kept one tab per exchanger, which meant one event had to be written in several places; the result was that the summary sheet missed 38 transactions worth 103,803,000 SDG and the exchanger tabs missed 8 worth 20,570,000. Excel read decimal numbers as dates. Account holders' names were spelled five ways.

What goes wrong, repeatedly:

- Payments cannot be matched to orders without reading a screenshot
- Proofs are not always forwarded to the exchanger
- Nobody knows how much room is left in an account today
- Stock leaves against a verbal confirmation
- Margin per shipment is known weeks after the goods have gone
- Nobody can prove the books balance

6 · Principles the system must hold to

These come from the owner's own rules for his administration. They are not preferences.

- **History must never move.** The pound falls almost daily, so rates change constantly. Every order line, receipt and conversion is stored with its amount, its currency and the rate of the day it happened, and that rate is never recalculated from a newer one. An order placed at 8,000 SDG per dollar still shows 8,000 in six months when the rate may be 11,000. A report on any past period must give exactly the same numbers whenever it is run, to the last pound. Money is stored as integers in the smallest unit, never as floating-point numbers.
- **Book per event, with accounts as balances.** Every money movement is one line with an origin and a destination. Customer to Umhani is one line; Umhani to an exchanger is a second; exchanger to the owner's euro account is a third. Never book a movement under its final destination and skip the intermediate step, or the intermediary becomes uncheckable.
- **The transaction code is unique.** The same code can appear only once. Duplicate screenshots are harmless because of it.
- **Same code, different amount: do not choose.** Show both to the owner.

- **When in doubt, book nothing.** An unknown account goes on a question list. Nothing is booked on it until the owner answers.
- **Show contradictions, never smooth them over.** A visible gap is better than a number that looks right.
- **Never invent an amount.** Mark every assumption.

7 · The build sequence

How the platform is built. One foundation, three modules on top of it, and a dashboard in front. Every feature we need belongs to exactly one of these. Nothing is built outside them.

Part	What it holds
Foundation	What every module stands on: login, users and roles, permissions in the database, environments for Shamsy and later for dealers, branding, and all configurable settings such as rates, currencies and discount thresholds.
Finance	Everything with money: orders, receipts and proofs, accounts and their daily limits, conversions between currencies, margin and profit, reports and closing checks.
Stock	Everything with goods: products, purchasing and shipments, landed cost, stock levels and movements, release against paid orders.
Communication	Everything with customers: customer records and history, quotes and price lists, and later WhatsApp.
Dashboard	The front layer. It holds no data of its own; it shows what the three modules hold, per role.

A marketing module may follow later; it is not part of this project.

Ranked by how much each step takes off our hands and how much money it stops us losing. Each step is a milestone and usable on its own. Steps 1 to 9 are this project. Steps 10 to 12 come later; they are listed so the foundation is built for them and nothing has to be done twice.

#	Step	Module	What it does and why
1	Platform setup	Foundation	Login: a personal account per user, email and password, with password reset. Sessions that survive a weak connection. No shared logins. Roles: four, each enforced in the database with row level security, not by hiding buttons. • Owner — everything: prices, rates, users, margins, approvals. • Marketing — reads customers, orders and stock, edits customers; never sees cost prices or margins. • Sales adviser — creates orders and receipts for her own customers; never sees cost prices or margins. • Warehouse — reads orders, records stock movements and releases; sees no financial data. Products and customers: the base records of the Stock and Communication modules, created in this step because every later step needs them. Account holders: every exchanger account with all its name spellings, so “Mogtaba”, “Mujtaba” and “Motgaba” are one person. Settings per environment: logo and house style, the discount thresholds (now 3% and 5%), the minimum rate, which currencies are used and which kinds of accounts exist — all configurable, none of them written into the code. Interface in English. Nothing else works without this step.
2	Orders	Finance	The adviser records an order once, as she does in the sales tool today: product and quantity, with totals in dollars and pounds. Prices: fixed in dollars; the adviser cannot change them. The owner can override a price manually. Exchange rate: the day's rate converts the dollar total to pounds. It can never be set below the minimum rate the owner has set. The rate says nothing about the price of a product; it only changes the currency. Discounts: a separate decision, per order line, independent of the rate. Every discount is highlighted — sand up to 3% of the line value, red from 3% to 5%, and above 5% it cannot be entered without the owner's approval. Payment instructions: the order tells the customer which account to pay into and in how many transfers of 3,000,000.
3	Receipts, with proof and status	Finance	Each receipt is one line keyed on its transaction code: date, amount, from which account, to which account, comment, photo of the proof. A receipt can pay several orders and an order can have several receipts. Duplicates are refused; the same code with a different amount is flagged. Each receipt carries a status: proof received, forwarded to the exchanger, confirmed by the exchanger. Anything stuck lights up. This is where our evenings go today, and where money disappears.

#	Step	Module	What it does and why
4	Accounts, limits and movements	Finance	Every exchanger account, the owner's own account, the pass-through accounts and the Airwallex account in the UAE, each with a live balance. Today's intake per account against the 15,000,000 daily limit, with a warning as it approaches. Transfers between accounts, payouts by exchangers in euros with their rate, cash taken by staff. Outstanding per exchanger. Pass-through accounts must stand at zero.
5	Release against paid orders	Stock	The warehouse can only release goods for an order that is fully paid. The smallest step, the largest drop in risk.
6	Purchasing, landed cost and stock	Stock	Euros converted to dollars with commission, giving the cost factor. Supplier invoices. Freight, customs in several payments at their own rates, inland transport and bank charges, in USD, EUR, AED or SDG, spread over the products pro rata to purchase value with a 20% safety uplift on indirect costs. On our current Growatt shipment that lifts the purchase price by 12.44%: an inverter bought at \$270 has a true cost of \$303.58. Stock on hand, minimum levels, every movement in and out, and what is in transit. Kits of several products sold as one, such as battery installation sets. Our own-brand model codes for OEM lines — SP for balance-of-system parts, SWF for pumps — alongside the supplier code.
7	Margin, profit and checks	Finance	Per product, order, customer, adviser, shipment, week and month — including a weekly overview — every result in dollars and euros , as an amount and as a margin: gross profit , net profit before currency , the currency result , and net profit after currency . The currency result per route — each exchanger and the UAE account — with the rate achieved and the volume converted. Money still held in dollars or dirham shown separately until it is converted. On our August 2026 shipment that read: revenue \$85,208, gross profit \$21,918 at 25.7%, a currency result of \$4,135, net profit \$27,604 at 30.3%. Expenses such as salaries, and refunds to customers, as their own lines. And the closing checks: calculated balance per account against the actual balance; sold equals received plus outstanding; received equals converted plus still on the accounts.
8	Quotes and price lists	Communication	A clean quote PDF in our house style, in English, with customer name, date, validity and payment terms such as 50% upfront and 50% before delivery. Some products, such as large solar pumps, are sold only to order against a deposit, never from stock. Sent through a WhatsApp share link. Status: sent, accepted, expired, and one click from an accepted quote to an order. The same catalogue produces the price list PDF.
9	CRM	Communication	Each dealer's profile: owner and decision-maker, phone and WhatsApp, city, type, purchasing power, monthly volume, brand preference, current supplier, and what he still has in stock from our last delivery. Segments A+ to D. Pipeline labels: consumer, trader, to verify, quote sent, paid. The source of each customer. Revenue per active dealer per month, and share of wallet: what else each dealer sells that he does not buy from us, at what price, from whom. That is the start of the market data this platform exists to collect.

#	Step	Module	What it does and why
10	WhatsApp Business API — later phase	Communication	Not in this project. A shared inbox on one WhatsApp number, every message linked to the customer record, approved templates for follow-ups and shipment announcements, and the 24-hour conversation window respected. Until then we use WhatsApp share links. Build the customer and message model so this plugs in later without rework.
11	Arabic interface — later phase	Foundation	Not in this project. An Arabic right-to-left interface and Arabic quote PDFs, as the advisers use today in the sales tool. Build for it from day one: no hard-coded text, every string in translation files, layouts that can be mirrored, number and date formats per language — so adding Arabic is translation work, not a rebuild.
12	Dealer access — later phase	Foundation	Not in this project. Dealers log in to run their own quotes, customers, stock and bookkeeping on our catalogue. Build the data model multi-tenant, with every permission in the database, from day one — so this becomes a switch rather than a rewrite. The platform is white-label: each dealer environment carries its own logo, colours and settings, so it can also be sold to others under their own brand.

Why receipts and accounts come before quotes: a quote saves an adviser a few minutes. Keying receipts on their code, tracking whether proofs reached the exchanger, and watching account limits save hours a day and stop money leaking. The quote calculation itself is already inside step 2; only the polished PDF waits until step 8.

8 · Screens, and what each one is for

Products and stock. Answer "do we have it" in two seconds on a phone. Record what came in and went out. Show what is below minimum and what is arriving. Cost price and margin exist but are invisible to advisers and the warehouse.

Customers. Search by name, city, type, adviser and segment. A dealer operating in two cities gets two records — deliberately.

Orders. Built like the sales tool: pick a customer, add products at fixed prices, set the rate, see totals in dollars and pounds. Payment instructions generated for the customer.

Receipts. The screen used most, on a phone, on a weak connection. Photo of the proof, transaction code, amount, accounts. Fast, and it must not lose data when the connection drops.

Accounts and exchangers. Every account with its balance and today's intake as a bar against the daily limit. Transfers and payouts. Outstanding per exchanger.

Open questions. Every movement the system could not place — unknown account, same code with different amounts, a payer who matches no customer — waiting for the owner's answer.

Dashboard. What the owner opens first. What needs attention today: accounts near their limit, proofs not forwarded, receipts not confirmed, customers overdue, stock below minimum, open questions. What this shipment earned: gross profit, net profit before currency, currency result and net profit after currency — each in dollars and euros, as amount and margin — plus the

currency result per conversion route. Where the cash is: at customers, at exchangers, in stock. How we are doing: margin over recent months, return on stock, and the share of orders delivered straight from stock.

Settings. Users and roles, the day's rate and the minimum rate — both changed by the owner, often daily — the calculation rate, the cost factor per shipment, reference data, company details.

9 · Where this is going, so build for it

Phase two turns the same system into a product our dealers use: their quotes, their customers, their stock, their bookkeeping, at a price that is easy to say yes to.

A dealer who runs his business inside our system orders from us. His quotes are built on our catalogue and our prices. And through it we see what the market is quoting and selling, and what our dealers still hold, weeks before it reaches us as an order. They get a system; we get the market.

So the data model is multi-tenant from day one, and every permission lives in the database. The platform is also **white-label and configurable**: each environment has its own branding, and every business rule that is specific to Shamsy — discount thresholds, minimum rate, currencies, account types — is a setting, not code. That keeps the door open to reselling it and to data-driven revenue beyond our own trade. Do not build the dealer screens. Do build a structure that will not need rewriting for them.

10 · The data model as we see it

A starting point, not a decision. Timestamps, soft delete and created_by everywhere. All money as integers in the smallest unit.

tenants — the company; Shamsy now, dealers later

users — name, role (owner / marketing / adviser / warehouse), language, prefix for order numbers

products — sku, name, brand, category (inverter / battery / panel / solar AC / pump / balance-of-system / accessory), purchase_price_usd, landed_cost_usd, list_price_usd, manual_price_usd, min_stock_level, active

kits and **kit_lines** — a bundle sold as one product

customers — name, type, city, phone, whatsapp, segment, source, assigned_adviser, owner_name, decision_maker, purchasing_power, monthly_volume, brand_preference, current_supplier, notes

orders and **order_lines** — order_number with adviser prefix, date, customer, adviser, rate_sdg_per_usd, status; lines with product, quantity, list price, discount, approved_by

quotes and **quote_lines** — as orders, plus validity, payment terms and status

accounts — holder, bank, account number, type (exchanger / own / pass-through / staff / euro / airwallex), exchanger, currency, daily_limit, active

account_aliases — every spelling of a holder's name

movements — the ledger. transaction_code (unique), date, amount, currency (USD / SDG / EUR / AED), rate, from_account, to_account, type (receipt / transfer / payout / purchase / expense / refund), comment, proof_url, proof_status (received / forwarded / confirmed)

movement_allocations — which part of a receipt pays which order; many to many

exchangers — name, contact, active

shipments, shipment_lines and **shipment_costs** — supplier, invoice, cost factor, and each cost with its own currency and rate

stock_movements — product, quantity, reason, order, shipment, released_by

exchange_rates — date, calculation rate, actual rates per exchanger and for the UAE account, set_by

conversions — date, from amount and currency, to amount and currency, rate, route (which exchanger or the UAE account), fee

open_questions — the movement or record, what is unclear, the answer, answered_by

11 · Non-functional requirements

- White-label and configurable: branding and business rules are settings per environment, never hard-coded.
- English interface in phase one. Arabic right-to-left follows later, so no hard-coded strings and a layout that can be mirrored.
- Works on a laptop and a phone browser. The team works on phones.
- Usable on 3G with frequent disconnects. Entries must survive a dropped connection.
- Numbers never parsed as dates, whatever the device locale.
- Every table paginated; assume tens of thousands of movements within two years.
- Audit trail on orders, movements and stock: who changed what, when.

Handover

A README covering local setup, environment variables and deployment. A schema diagram. A seed script with realistic data from our real files. A short walkthrough of each screen. And a one-page work instruction for advisers and the owner: what an adviser records at the moment of sale, how a proof is numbered and filed, who forwards it to the exchanger and how that is ticked off, and what the owner checks daily and weekly.

What we supply on day one

- The sales tool, Shamsy_Verkooptool_v7.html, version 7
- The daily administration, Shamsy_Daily_Admin_v10.xlsx
- The shipment dossier, Shamsy_Zending_Growatt_v24.xlsx
- The cash register built from 211 payment screenshots
- A sample of those screenshots, to see the bankak format

- The HTML dashboard prototype, with layout, colours and typography
- The price list PDF
- Brand assets: logos with and without the Arabic wordmark, colours, fonts
- Access to our GitHub organisation and Supabase project

How we build

- **Technical lead.** Every feature is defined in advance by our technical lead, with a goal prompt, acceptance tests and the functional result expected. A feature is finished when its tests pass and it works as described.
- **Code.** Next.js with TypeScript in an Nx monorepo on GitHub; Supabase; Tailwind; Vercel.
- **Tests and deployment.** GitHub Actions run the tests and deploy on every change. Every change comes with tests and goes in through a reviewed pull request.
- **Documentation** is written with each feature, not at the end.

12 · Going live

The system does not start empty. Before the team switches over, the developer imports from our spreadsheets:

- the opening balance of every account, as confirmed by the owner on the switch-over date
- what every customer still owes, and every open order
- current stock per product, with its landed cost
- the ledger rebuilt from the payment screenshots in the cash register, with its open questions

After go-live the spreadsheets stop. Running both side by side is how the old books went wrong.

13 · Acceptance tests

A milestone is paid when its tests pass on the live system. The minimum set:

- **Landed cost.** The current Growatt shipment's costs, entered as they are, give an inverter bought at \$270 a cost price of \$303.58.
- **Margin.** The August 2026 shipment gives, in dollars: revenue \$85,208, gross profit \$21,918 at 25.7%, a currency result of \$4,135, and net profit after currency of \$27,604 at 30.3%. The same lines in euros follow from the cost factor and the actual payouts recorded, and switching between dollars and euros never changes the underlying records.
- **History.** Run last month's report, change today's rate, run it again: every number is identical, to the last pound.
- **Permissions.** Logged in as an adviser or as marketing, a cost price or margin cannot be retrieved — not on screen, not through the API, not by querying the database.
- **Duplicates.** Entering the same transaction code twice is refused; the same code with a different amount lands on the open-questions list.
- **Daily limit.** Receipts on one account that pass 15,000,000 SDG on the same Sudanese day trigger the warning.
- **Discounts.** 2% shows sand, 4% shows red, 6% cannot be saved without the owner's approval.

- **Release.** The warehouse cannot release an order that is not fully paid.

14 · Running it safely

- **Hosting.** Supabase on a paid plan, not the free tier, which pauses when idle. Daily backups with point-in-time recovery. Paid by us, on our account.
- **Export.** A full export of all data at any time, so we are never locked in.
- **Owner account.** Two-factor authentication on the owner's login. That account approves discounts and sees everything.
- **Time zone.** The business day is Sudan time (Africa/Khartoum). The daily account limit resets at midnight in Sudan, whatever time zone the owner is working in. Store timestamps in UTC and display them in the user's own time zone.
- **Warranty.** Sixty days of free fixes after final handover, for anything that does not work as specified.
- **Confidentiality.** This brief, our data and our screenshots are shared only after a signed confidentiality agreement.

15 · Paid trial task

What this is. Before we award the project, the shortlisted candidates each build one small, real piece of it: the screen where a sales adviser records an order. It takes six to eight hours and we pay **\$100** for it, whether or not we continue with you. It shows us how you think about money, rules and phones — which is what the whole project depends on.

The situation you are building for. A dealer in Khartoum asks our sales adviser on WhatsApp for four inverters, two small batteries and one large battery. The adviser opens this screen on her phone, picks the dealer, adds the products, gives a small discount where she is allowed to, enters today's exchange rate, and sees what the dealer has to pay in US dollars and in Sudanese pounds. Then she saves the order.

The rules the screen must enforce

1. **Prices are fixed.** Each product has a price in US dollars. The adviser cannot change it.
2. **Discount is per line, and separate from everything else.** The adviser may give a discount in dollars on any line. The discount is measured as a percentage of that line's value (price × quantity):
 - above 0% and up to 3% — the line turns **sand**, and saving is allowed
 - above 3% and up to 5% — the line turns **red**, and saving is still allowed
 - above 5% — saving is **blocked** until the owner approves that line
3. **The exchange rate is one number for the whole order.** It converts dollars to pounds. It has nothing to do with discounts or prices. It can never be lower than the **minimum rate of 8,000 SDG per dollar**. If the adviser types a lower number, the screen refuses it and puts the value back to 8,000. A higher number is always fine.
4. **A saved order never changes.** The rate the adviser entered is stored on the order. If the rate setting changes tomorrow, yesterday's order still shows exactly the same amounts.

Test data to use

Product	Price per unit
SPF 6000 ES Plus — 6 kW inverter	\$515
SPE 12000 ES — 12 kW inverter	\$975
Hope 5.0L-B1 — 5 kWh battery	\$810
Hope 16.0LM-A1 — 16 kWh battery	\$2,070

Customers: three invented dealers, for example Ahmed Trading in Khartoum, Nile Solar in Omdurman and Dongola Power in Dongola.

Users: two accounts — one **adviser**, one **owner**. Only the owner can approve a discount above 5%.

A worked example — your screen must produce exactly these numbers

Rate entered: 8,200 SDG per dollar.

Line	Quantity × price	Line value	Discount	Discount %	Colour	Line total
SPF 6000 ES Plus	4 × \$515	\$2,060	\$40	1.94%	sand	\$2,020
Hope 5.0L-B1	2 × \$810	\$1,620	\$70	4.32%	red	\$1,550
Hope 16.0LM-A1	1 × \$2,070	\$2,070	\$150	7.25%	blocked	\$1,920

As the adviser, the order cannot be saved while line 3 is blocked. If she removes line 3 instead, the order totals **\$3,570**, which is **29,274,000 SDG**, and can be saved.

After the owner approves line 3: the order totals **\$5,490**, which is **45,018,000 SDG**.

Then change the rate setting to 9,000 and reopen the saved order. It must still show 8,200 and 45,018,000 SDG.

Try to enter a rate of 7,900: it must be refused and return to 8,000.

What you do not need to build. Payments, stock, PDF quotes, customer management, Arabic, and polished design. Plain and clean is fine. We are judging the logic, not the looks.

Technical expectations

- Next.js with TypeScript, Supabase, Tailwind — the same stack as the real project
- Money stored as whole numbers in cents, never as decimals
- The rate stored on each saved order
- The discount rules enforced in the database or on the server, not only in the browser
- Works on a phone screen

What you deliver

1. A link to the working screen, hosted on your own Vercel and Supabase

2. Access to the code repository
3. A screen recording of five minutes or less, walking through the worked example above
4. A short note: what you would do differently when building the real system, and anything in our rules that seems unclear or wrong

How we judge it

What we check	Why it matters
The worked example gives exactly the numbers above	Every report in the real system depends on correct arithmetic
The 5% block cannot be bypassed, not even by calling the server directly	Discount control is money control
A saved order does not change when the rate changes	This is the rule the whole administration rests on
Money is stored in cents and the rate is stored on the order	Rounding errors and moving history are the costliest bugs we can get
It works on a phone	Our advisers work on phones, on weak connections
Your note shows you understood the problem	We want someone who thinks along, not only builds

Time and payment. Six to eight hours. You have three working days from receiving the task. We pay **\$100** on delivery, whether or not we award you the project.

16 · What we are not asking for, yet

The WhatsApp Business API with a shared inbox — share links do the job for now. Dealer logins and dealer screens — build the model, not the screens. A native mobile app. Accounting export. Automatic reading of payment screenshots. An Arabic right-to-left interface and Arabic quotes — build for them, do not build them yet. Warranty and serial-number tracking — though we will need warranty and return handling before our own-brand OEM lines go on sale, so design the product model with it in mind.

If you think one of these belongs in phase one, tell us why.

17 · Where we want your opinion

We are traders, not software people. Tell us where we are wrong.

- Our receipts arrive as Arabic bankak screenshots. Would you extract the transaction code, amount, accounts and comment automatically from the image, and at what point would that pay for itself? We have 211 of them from one shipment.
- A receipt can pay several orders. How would you let an adviser allocate it quickly on a phone?
- We plan to store the rate on each order and each movement. Is there a better way to keep reports stable over time?
- The sales tool works offline today. How much of that do we lose, and how would you keep order entry working on a dropped connection?

- What would you change in the data model to make dealer access a switch rather than a rewrite?
- What are we going to regret in six months that we cannot see now?